COMICCRAFT — CODING & SOLUTION

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KEY CODE SNIPPETS

1. OUTLINE GENERATION (app/gemini_outline.py)

```
def generate_outline(user_prompt):
    prompt = f"""Generate a JSON array of 5 comic panels for: {user_prompt}
    Each object must have:
    - "panel" (1-5)
    - "title" (short)
    - "scene_description" (1-2 sentences)
    - "image_prompt" (detailed visual for image generation)
    Return ONLY the JSON array."""
    response = client.models.generate_content(
        model=MODEL_NAME, contents=prompt)
    raw = re.sub(r"```json|```", "", response.text.strip()).strip()
    match = re.search(r"\[.*\]", raw, re.DOTALL)
    if match:
        raw = match.group(0)
    return json.loads(raw)
```

2. STORY GENERATION (app/gemini_story.py)

```
def generate_story(panels):
```

```
    outline = "\n".join([
        f"Panel {p.get('panel')}: {p.get('title')} — {p.get('scene_description')}"
        for p in panels])
    prompt = f"""Write a comic-style narration for these panels.
    Format each panel as "**Panel X:**" so it can be split.
    {outline}"""
    response = client.models.generate_content(
        model=MODEL_NAME, contents=prompt)
    return response.text.strip()
```

3. IMAGE GENERATION (app/image_generator.py)

```
def generate_image(prompt, width=768, height=768):
    os.makedirs("static/panels", exist_ok=True)
    filepath = os.path.join("static", "panels", f"{uuid.uuid4().hex}.png")
    url =
f"https://image.pollinations.ai/prompt/{quote(prompt)}?width={width}&height={height}&no
logo=true"
    r = requests.get(url, timeout=120)
    r.raise_for_status()
    with open(filepath, "wb") as f:
        f.write(r.content)
    return filepath.replace("\\", "/")
```

4. MAIN ROUTE (app/main.py)

```
@app.post("/generate", response_class=HTMLResponse)
```

```
async def generate(request: Request,
                   prompt: str = Form(...),
                   character_name: str = Form(...),
                   setting: str = Form(...),
                   tone: str = Form(...),
                   style: str = Form(...)):
    full_prompt = f"{prompt}. Character: {character_name}. " \
                  f"Setting: {setting}. Tone: {tone}. Style: {style}."
    panels = generate_outline(full_prompt)
    story = generate_story(panels)
    chunks = split_story_by_panel(story, len(panels))
    layout = []
    for i, p in enumerate(panels):
        img = generate_image(p.get("image_prompt", p.get("scene_description")))
        layout.append({
            "panel": p.get("panel", i+1),
            "title": p.get("title"),
            "scene_description": p.get("scene_description"),
            "image_path": "/" + img if img else None,
            "text": chunks[i] if i < len(chunks) else story
        })
    return templates.TemplateResponse(
        request=request, name="comic_preview.html",
        context={"layout": layout, "error": None})
```